

Functions of One Real Variable (2005–2014)

Category: Real Analysis • Official Mock Test Series

TOTAL QUESTIONS

25

TOTAL MARKS

111 M

TEST DURATION

75 Min

PAPER PATTERN

Classic Paper Pattern (2

Section / Type		Questions	Marking Scheme	Negative Marking	Format / Details
Multiple Choice (MCQ)		25	+1 / +2 Marks	−1/3 / −2/3 Marks	4 Options (1 Correct)

IMPORTANT INSTRUCTIONS FOR CANDIDATES

- Authentic Examination PYQs:** This mock paper contains verified official IIT JAM Mathematics questions and high-resolution problem statements.
- Section A (MCQ):** Each question has four choices (A), (B), (C), (D), out of which **only one** is correct. Wrong answers carry negative penalty (1/3 of positive marks).
- Section B (MSQ):** Each question has four choices (A), (B), (C), (D), out of which **one or more than one** choice(s) may be correct. Full marks are awarded only if all correct choices and zero incorrect choices are chosen. No partial marks and no negative marks.
- Section C (NAT):** Numerical answers are real numbers entered via virtual keyboard. Full credit is awarded if the numerical answer falls within the official accepted interval. No negative marking.
- Master Answer Key:** Complete official answers and acceptance bounds are provided at the end of this document.
- Online Interactive Simulator:** Practice this test with virtual calculator, timers, and instant scoring analytics at vaibhavgeometer.github.io.

IIT JAM Mathematics Mock Test Series • Curated by Vaibhav Geometer

Question 1 (JAM 2014 • Q33 • ID: JAM_2014_Q33)

MCQ

+2 M

Neg: -0.67

Q.33 Let $f : (0, \infty) \rightarrow \mathbb{R}$ be a differentiable function such that $f'(x^2) = 1 - x^3$ for all $x > 0$ and $f(1) = 0$. Then $f(4)$ equals

- (A) $\frac{-47}{5}$ (B) $\frac{-47}{10}$ (C) $\frac{-16}{5}$ (D) $\frac{-8}{5}$

Question 2 (JAM 2014 • Q26 • ID: JAM_2014_Q26)

MCQ

+2 M

Neg: -0.67

Q.26 Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function with continuous derivative such that $f(\sqrt{2}) = 2$ and $f(x) = \lim_{t \rightarrow 0} \frac{1}{2t} \int_{x-t}^{x+t} s f'(s) ds$ for all $x \in \mathbb{R}$. Then $f(3)$ equals

- (A) $\sqrt{3}$ (B) $3\sqrt{2}$ (C) $3\sqrt{3}$ (D) 9

Question 3 (JAM 2014 • Q23 • ID: JAM_2014_Q23)

MCQ

+2 M

Neg: -0.67

Q.23 Let $f(x) = x^3 + x$ and $g(x) = x^3 - x$ for all $x \in \mathbb{R}$. If f^{-1} denotes the inverse function of f , then the derivative of the composite function $g \circ f^{-1}$ at the point 2 is

- (A) $\frac{2}{13}$ (B) $\frac{1}{2}$ (C) $\frac{11}{13}$ (D) $\frac{11}{4}$

Question 4 (JAM 2014 • Q22 • ID: JAM_2014_Q22)

MCQ

+2 M

Neg: -0.67

Q.22 Let $S = [0, 1] \cup [2, 3)$ and let $f : S \rightarrow \mathbb{R}$ be defined by $f(x) = \begin{cases} 2x & \text{if } x \in [0, 1], \\ 8 - 2x & \text{if } x \in [2, 3). \end{cases}$

If $T = \{f(x) : x \in S\}$, then the inverse function $f^{-1} : T \rightarrow S$

- (A) does NOT exist (B) exists and is continuous
(C) exists and is NOT continuous (D) exists and is monotonic

Question 5 (JAM 2014 • Q10 • ID: JAM_2014_Q10)

MCQ

+1 M

Neg: -0.33

Q.10 Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function satisfying $x + \int_0^x f(t) dt = e^x - 1$ for all $x \in \mathbb{R}$. Then the set $\{x \in \mathbb{R} : 1 \leq f(x) \leq 2\}$ is the interval

(A) $[\log 2, \log 3]$ (B) $[2 \log 2, 3 \log 3]$ (C) $[e-1, e^2-1]$ (D) $[0, e^2]$

Question 6 (JAM 2014 • Q2 • ID: JAM_2014_Q2)

MCQ

+1 M

Neg: -0.33

Q.2 The coefficient of $(x-1)^2$ in the Taylor series expansion of $f(x) = xe^x$ ($x \in \mathbb{R}$) about the point $x=1$ is

(A) $\frac{e}{2}$ (B) $2e$ (C) $\frac{3e}{2}$ (D) $3e$

Question 7 (JAM 2014 • Q1 • ID: JAM_2014_Q1)

MCQ

+1 M

Neg: -0.33

Q.1 Let $f(x) = |x^2 - 25|$ for all $x \in \mathbb{R}$. The total number of points of $\mathbb{R}$ at which f attains a local extremum (minimum or maximum) is

(A) 1

(B) 2

(C) 3

(D) 4

Question 8 (JAM 2013 • Q5 • ID: JAM_2013_Q5)

MCQ

+2 M

Neg: -0.67

Let the function $f(x)$ be defined by

$$f(x) = \begin{cases} e^x, & x \text{ is rational} \\ e^{1-x}, & x \text{ is irrational} \end{cases}$$

for x in $(0, 1)$. Then

(A) f is continuous at every point in $(0, 1)$ (B) f is discontinuous at every point in $(0, 1)$ (C) f is discontinuous only at one point in $(0, 1)$ (D) f is continuous only at one point in $(0, 1)$

Question 9 (JAM 2013 • Q4 • ID: JAM_2013_Q4)

MCQ

+2 M

Neg: -0.67

Q.4 Let $[x]$ denote the greatest integer function of x . The value of α for which the function

$$f(x) = \begin{cases} \frac{\sin[-x^2]}{[-x^2]}, & x \neq 0 \\ \alpha, & x = 0 \end{cases}$$

is continuous at $x=0$ is

- (A) 0 (B) $\sin(-1)$ (C) $\sin 1$ (D) 1

Question 10 (JAM 2012 • Q2 • ID: JAM_2012_Q2)

MCQ

+6 M

Neg: -2.0

Q.2 The number of distinct real roots of the equation $x^9 + x^7 + x^5 + x^3 + x + 1 = 0$ is

- (A) 1 (B) 3 (C) 5 (D) 9

Question 11 (JAM 2011 • Q2 • ID: JAM_2011_Q2)

MCQ

+6 M

Neg: -2.0

Q.2 The number of real roots of the equation $x^3 + x - 1 = 0$ is

- (A) 0 (B) 1 (C) 2 (D) 3

Question 12 (JAM 2010 • Q5 • ID: JAM_2010_Q5)

MCQ

+6 M

Neg: -2.0

Q.5 Let $f: (1,4) \rightarrow \mathbf{R}$ be a uniformly continuous function and let $\{a_n\}$ be a Cauchy sequence in $(1,2)$. Let $x_n = a_n^2 f(a_n^2)$ and $y_n = \frac{1}{1+a_n^2} f(a_n^2)$, for all $n \in \mathbf{N}$. Which of the following statements is true?

- (A) Both $\{x_n\}$ and $\{y_n\}$ must be Cauchy sequences in $\mathbf{R}$
 (B) $\{x_n\}$ must be a Cauchy sequence in $\mathbf{R}$ but $\{y_n\}$ need not be a Cauchy sequence in $\mathbf{R}$
 (C) $\{y_n\}$ must be a Cauchy sequence in $\mathbf{R}$ but $\{x_n\}$ need not be a Cauchy sequence in $\mathbf{R}$
 (D) Neither $\{x_n\}$ nor $\{y_n\}$ needs to be a Cauchy sequence in $\mathbf{R}$

Question 13 (JAM 2009 • Q10 • ID: JAM_2009_Q10)

MCQ

+6 M

Neg: -2.0

Q.10 Let a be a non-zero real number. Then $\lim_{x \rightarrow a} \frac{1}{x^2 - a^2} \int_a^x \sin(t^2) dt$ equals

- (A) $\frac{1}{2a} \sin(a^2)$ (B) $\frac{1}{2a} \cos(a^2)$ (C) $-\frac{1}{2a} \sin(a^2)$ (D) $-\frac{1}{2a} \cos(a^2)$

Question 14 (JAM 2009 • Q3 • ID: JAM_2009_Q3)

MCQ

+6 M

Neg: -2.0

Q.3 Which of the following sequences of functions is uniformly convergent on $(0, 1)$?

- (A) x^n (B) $\frac{n}{nx+1}$ (C) $\frac{x}{nx+1}$ (D) $\frac{1}{nx+1}$

Question 15 (JAM 2008 • Q13 • ID: JAM_2008_Q13)

MCQ

+6 M

Neg: -2.0

Q.13 Let $f: (-1, 1) \rightarrow \mathbb{R}$ be defined as $f(x) = \frac{x^2}{1 - \cos x}$ for $x \neq 0$ and $f(0) = 2$. If

$f(x) = \sum_{n=0}^{\infty} a_n x^n$ is the Taylor expansion of f for all x in $(-1, 1)$, then $\sum_{n=0}^{\infty} a_{2n+1}$ is

- (A) 0 (B) $1/2$ (C) 1 (D) 2

Question 16 (JAM 2008 • Q8 • ID: JAM_2008_Q8)

MCQ

+6 M

Neg: -2.0

Q.8 Let $f(x) = \int_0^x (x^2 - t^2) g(t) dt$, where g is a real valued continuous function on $\mathbb{R}$. Then $f'(x)$ is equal to

- (A) 0 (B) $x^3 g(x)$ (C) $\int_0^x g(t) dt$ (D) $2x \int_0^x g(t) dt$

Question 17 (JAM 2008 • Q6 • ID: JAM_2008_Q6)

MCQ

+6 M

Neg: -2.0

Q.6

Let $V = \{(x, y, z) \in \mathbb{R}^3 : \frac{1}{4} \leq x^2 + y^2 + z^2 \leq 1\}$ and $\vec{F} = \frac{x\hat{i} + y\hat{j} + z\hat{k}}{(x^2 + y^2 + z^2)^2}$ for $(x, y, z) \in V$.

Let $\hat{n}$ denote the outward unit normal vector to the boundary of V and S denote the part $\{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 + z^2 = \frac{1}{4}\}$ of the boundary of V . Then $\iint_S \vec{F} \cdot \hat{n} \, dS$ is equal to

(A) -8π (B) -4π (C) 4π (D) 8π

Question 18 (JAM 2007 • Q12 • ID: JAM_2007_Q12)

MCQ

+6 M

Neg: -2.0

12. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function. If $\int_0^x f(2t) dt = \frac{x}{\pi} \sin(\pi x)$ for all $x \in \mathbb{R}$, then $f(2)$ is equal to

(A) -1 (B) 0 (C) 1 (D) 2

Question 19 (JAM 2007 • Q8 • ID: JAM_2007_Q8)

MCQ

+6 M

Neg: -2.0

8. For every function $f: [0, 1] \rightarrow \mathbb{R}$ which is twice differentiable and satisfies $f'(x) \geq 1$ for all $x \in [0, 1]$, we must have

(A) $f''(x) \geq 0$ for all $x \in [0, 1]$ (B) $f(x) \geq x$ for all $x \in [0, 1]$ (C) $f(x_2) - x_2 \leq f(x_1) - x_1$ for all $x_1, x_2 \in [0, 1]$ with $x_2 \geq x_1$ (D) $f(x_2) - x_2 \geq f(x_1) - x_1$ for all $x_1, x_2 \in [0, 1]$ with $x_2 \geq x_1$

Question 20 (JAM 2007 • Q3 • ID: JAM_2007_Q3)

MCQ

+6 M

Neg: -2.0

3. Which of the following functions is uniformly continuous on the domain as stated?

(A) $f(x) = x^2, \quad x \in \mathbf{R}$

(B) $f(x) = \frac{1}{x}, \quad x \in [1, \infty)$

(C) $f(x) = \tan x, \quad x \in (-\pi/2, \pi/2)$

(D) $f(x) = [x], \quad x \in [0, 1]$

($[x]$ is the greatest integer less than or equal to x)

Question 21 (JAM 2007 • Q1 • ID: JAM_2007_Q1)

MCQ

+6 M

Neg: -2.0

1. Let $A(t)$ denote the area bounded by the curve $y = e^{-|x|}$, the x -axis and the straight lines $x = -t$ and $x = t$. Then $\lim_{t \rightarrow \infty} A(t)$ is equal to

(A) 2

(B) 1

(C) 1/2

(D) 0

Question 22 (JAM 2006 • Q15 • ID: JAM_2006_Q15)

MCQ

+6 M

Neg: -2.0

15. Let $f_n(x) = n \sin^{2n+1} x \cos x$. Then the value of

$$\lim_{n \rightarrow \infty} \int_0^{\pi/2} f_n(x) dx - \int_0^{\pi/2} \left(\lim_{n \rightarrow \infty} f_n(x) \right) dx$$

is

(A) 1/2

(B) 0

(C) -1/2

(D) $-\infty$

Question 23 (JAM 2006 • Q5 • ID: JAM_2006_Q5)

MCQ

+6 M

Neg: -2.0

5. Let $f(x) = \int_{\sin x}^{\cos x} e^{-t^2} dt$. Then $f'(\pi/4)$ equals
- (A) $\sqrt{1/e}$
(B) $-\sqrt{2/e}$
(C) $\sqrt{2/e}$
(D) $-\sqrt{1/e}$

Question 24 (JAM 2006 • Q2 • ID: JAM_2006_Q2)

MCQ

+6 M

Neg: -2.0

2. Let $f(x) = (x-2)^{17}(x+5)^{24}$. Then
- (A) f does not have a critical point at 2
(B) f has a minimum at 2
(C) f has a maximum at 2
(D) f has neither a minimum nor a maximum at 2

Question 25 (JAM 2005 • Q15 • ID: JAM_2005_Q15)

MCQ

+6 M

Neg: -2.0

15. Let $a > 1$ and $f, g, h : [-a, a] \rightarrow \mathbf{R}$ be twice differentiable functions such that for some c with $0 < c < 1 < a$,

$$f(x) = 0 \text{ only for } x = -a, 0, a;$$

$$f'(x) = 0 = g(x) \text{ only } x = -1, 0, 1;$$

$$g'(x) = 0 = h(x) \text{ only for } x = -c, c.$$

The possible relations between f, g, h are

- (A) $f = g'$ and $h = f'$
(B) $f' = g$ and $g' = h$
(C) $f = -g'$ and $h' = g$
(D) $f = -g'$ and $h' = f$

OFFICIAL MASTER ANSWER KEY & EVALUATION RUBRIC

Functions of One Real Variable (2005–2014) • All Official Answers & Acceptance Ranges

Q. No.	Type	Marks	Official Key
1	MCQ	+2	A
2	MCQ	+2	B
3	MCQ	+2	B
4	MCQ	+2	C
5	MCQ	+1	A
6	MCQ	+1	C
7	MCQ	+1	C
8	MCQ	+2	D
9	MCQ	+2	C
10	MCQ	+6	A
11	MCQ	+6	B
12	MCQ	+6	A
13	MCQ	+6	A

Q. No.	Type	Marks	Official Key
14	MCQ	+6	C
15	MCQ	+6	A
16	MCQ	+6	D
17	MCQ	+6	A
18	MCQ	+6	A
19	MCQ	+6	D
20	MCQ	+6	B
21	MCQ	+6	A
22	MCQ	+6	A
23	MCQ	+6	B
24	MCQ	+6	D
25	MCQ	+6	B

SCORING METHODOLOGY & EVALUATION GUIDELINES

- **Multiple Choice Questions (MCQ):** Full marks for correct single choice. Negative marks deducted for each incorrect attempt ($-1/3$ for 1-mark questions, $-2/3$ for 2-mark questions). Unattempted questions award zero.
- **Multiple Select Questions (MSQ):** Full marks if and only if all correct choices are marked and no incorrect choice is selected. Zero marks for partial selection or any incorrect choice. No negative marking.
- **Numerical Answer Type (NAT):** Any numerical value falling strictly within the specified official range (e.g., $[a, b]$) receives full credit. No negative marking.
- **Marks to All (MTA):** Questions officially designated as MTA award full marks to all candidates regardless of attempt.